



## The Problem

Modern AI is hitting the Silicon Wall. Dense transformers require millions in compute and kW-scale power. Biological brains run on ~20W. GPT-3 inference uses ~600J per token. The scaling laws are breaking. Standard LLMs cannot reason about sparse, event-driven biological or neuromorphic systems.

## Grounding Reality

Eon Systems recently emulated 125,000 biological neurons to drive a simulated fly body in real-time. This validates that structural wiring alone can generate complex behavior. SynapTechBio is the compiler that commercializes this insight for enterprise deployment.

## The Solution: IDRE

The Integrated Data Representation Engine (IDRE) is a GPU-accelerated sparse graph engine executing the Drosophila brain connectome (139k neurons, 50M synapses) in real-time. 5x compute latency reduction and 20x energy savings vs dense architectures. ~1ms activation on GPU. Real-time 3D brain visualization. Intel Loihi bridge. Data Capture Split Layer for cryptographic IP protection.

## Scientific Validation

FlyWire Connectome: Dorkenwald et al., Nature 2024 - first complete adult brain wiring diagram. 139,255 neurons, ~50M synapses, CC BY 4.0. 50+ neuroscientists validated. FlyWire LSM: >95% training accuracy on next-token prediction. Trains in <30 seconds on CPU.

## Business Model: RPaaS

Research-Platform-as-a-Service. \$300k ACV per clinical hub. Target: 1,500 hubs in 5 years = \$450M SOM. TAM: \$1.8T (Global AI Infrastructure) | SAM: \$300B (Next-Gen Cloud & Neuromorphic)

## Competitive Advantage

Only platform combining open connectome access, cloud Loihi hardware, live graph computation, IP-protected research via DCSL, and a continuously improving open-weights foundation model (the Living Blueprint). No competitor occupies this space.

## Ask: \$150,000 Pre-Seed

60% Compute Infrastructure (Cloud GPU, Loihi access, infra) | 40% Engineering & IP (2 FTE, 6 months + legal). Delaware C-Corp, post-money SAFE. 18-month runway. Key milestones: security audit, first researcher onboarded, YC/NEA product demo, 10k queries/day.

## Due Diligence Highlights

Connectome-Function: IDRE solves structural routing, not consciousness. Biological inductive biases for functional efficiency.

DCSL Moat: Cryptographic fork encrypts IP while creating a non-scrappable proprietary dataset.

Hardware-Agnostic: CuPy to SciPy to NumPy fallback. No single point of hardware failure.

Funding Frame: \$150k is a Technical Phase Bridge to 4 patents + Fruit Fly Benchmark + producible demo.

## Contact:

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15-Minute Introductory Sync Available